

Modelling and Simulation

Lecture 3: Dynamical systems and numerical integration

1. General introduction to dynamical systems

What is a dynamical system?

- A system that varies through time.
- The system is described by its state $s(t)$.
- The state $\mathbf{s}$ can be multivariate (a vector)

$$\mathbf{s}(t) = (s_1(t), \dots, s_n(t))^T.$$

- Examples: whether evolution, population evolution, ...

Different classes of descriptions

- **Discrete** dynamical systems are described by *recurrence relations*

$$s(t + \Delta t) = f(s(t))$$

where f is a function and Δt the time increment.

- **Continuous** dynamical systems are described by *differential equations*

$$\dot{s} \equiv \frac{ds}{dt} = f(s(t))$$

Generality of the description: continuous case

What happens if we have second derivatives?

$$\dot{s} + \ddot{s} = f(s)$$

Let us then define

$$\dot{s} = y$$

$$\dot{y} = f(s) - y$$

By defining $\dot{\mathbf{u}}(t) \equiv (s(t), y(t))$ we have a system of the form

$$\dot{\mathbf{u}} = g(\mathbf{u})$$

Generality of the description: discrete case

“Second derivatives” equivalent to presence of a $t - \Delta t$ term

$$s(t + \Delta t) = f(s(t)) + s(t - \Delta t).$$

Similarly to continuous case

$$y(t + \Delta t) = s(t),$$

$$s(t + \Delta t) = f(s(t)) + y(t),$$

and by defining $\mathbf{u}(t) \equiv (s(t), y(t))$ we have a system of the form

$$\mathbf{u}(t + \Delta t) = g(\mathbf{u}(t)).$$

Generality of the description: continuous case

What happens if f explicitly depends on t ?

$$\dot{s} = f(s, t)$$

Let us then define

$$\dot{s} = f(s, y)$$

$$\dot{y} = 1,$$

with $y(0) = 0$. By defining $\mathbf{u}(t) \equiv (s(t), y(t))$ we have a system of the form

$$\dot{\mathbf{u}} = g(\mathbf{u})$$

2. The random walk

The random walk

Time evolution of particles, drunk people, ...

$$x(t + \Delta t) = x(t) + \Delta t \cdot v_x(t),$$

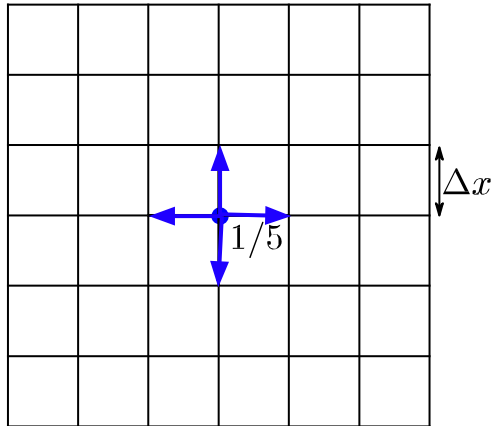
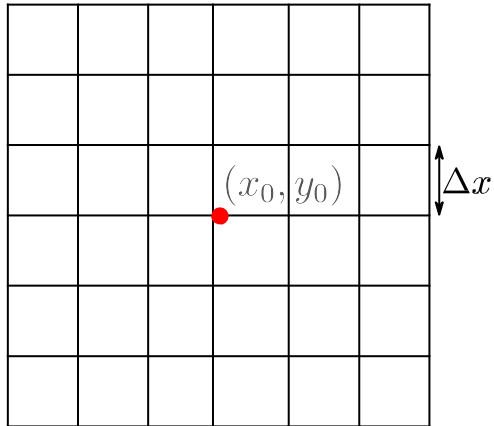
$$y(t + \Delta t) = y(t) + \Delta t \cdot v_y(t),$$

where (v_x, v_y) is chosen randomly in the ensemble

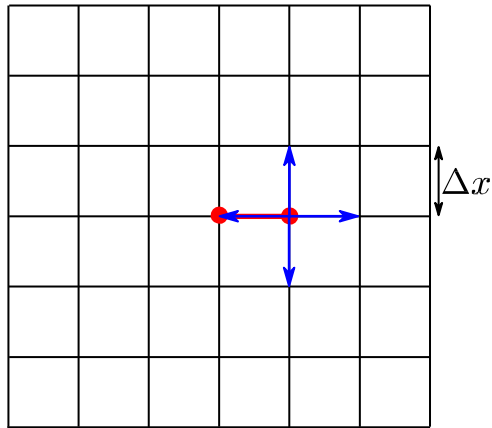
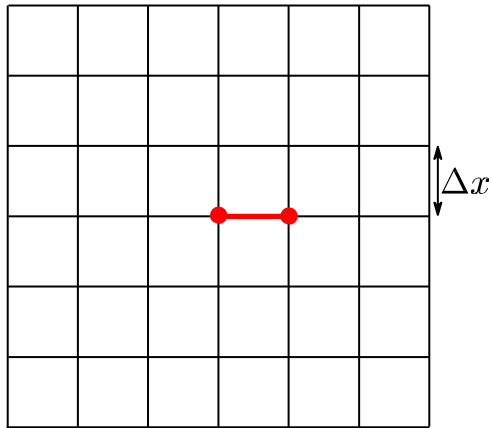
$$\{(\Delta x/\Delta t, 0), (-\Delta x/\Delta t, 0), (0, 0), (0, \Delta y/\Delta t), (0, -\Delta y/\Delta t)\},$$

where Δx and Δy are the length of the displacements in direction x and y .

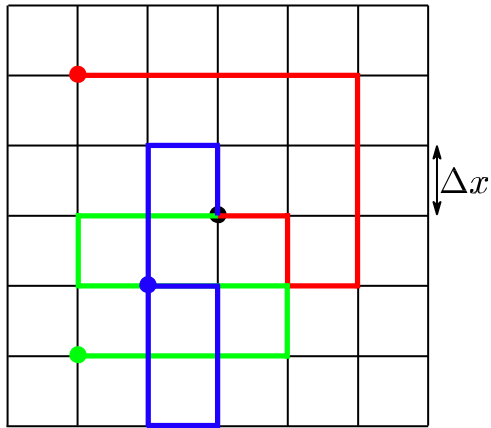
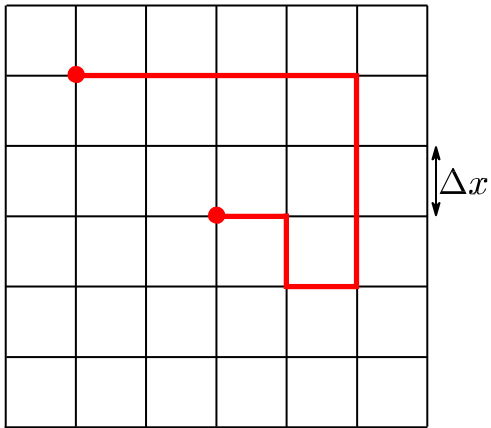
The random walk: illustration



The random walk: illustration



The random walk: illustration



The random walk: mathematical analysis

What is the “average” behavior of many random walkers?

- Initial position of the walker is $(x(0), y(0)) = (x_0, y_0)$.
- After a time $t = N \cdot \Delta t$ the position of a random walker is

$$\begin{aligned}x(t) &= x(t - \Delta t) + \Delta t \cdot v_x(t - \Delta t) \\&= x(t - 2\Delta t) + \Delta t \cdot (v_x(t - 2\Delta t) + v_x(t - \Delta t)), \\&= x(0) + \Delta t \sum_{n=1}^N v_x(t - n \cdot \Delta t).\end{aligned}$$

The random walk: mathematical analysis

What is the average position?

$$\langle x(t) \rangle = x_0 + \Delta t \sum_{n=1}^N \langle v_x(t - n \cdot \Delta t) \rangle$$

where for any time, τ , one has

$$\langle v_x(\tau) \rangle = \frac{1}{5} \left(-\frac{\Delta x}{\Delta t} \right) + \frac{3}{5} \cdot 0 + \frac{1}{5} \frac{\Delta x}{\Delta t} = 0$$

and therefore

$$\langle x(t) \rangle = x_0$$

The random walk: mathematical analysis

What is the average displacement?

$$d(t) = \sqrt{\langle (x(t) - x_0)^2 \rangle + \langle (y(t) - y_0)^2 \rangle}$$

Therefore with $d_x^2 = \langle (x(t) - x_0)^2 \rangle$

$$\begin{aligned} d_x^2 &= \Delta t^2 \left\langle \sum_{m=1}^N v_x(t - m\Delta t) \sum_{n=1}^N v_x(t - n\Delta t) \right\rangle \\ &= \Delta t^2 \sum_{m,n=1}^N \langle v_x(t - m\Delta t) v_x(t - n\Delta t) \rangle \end{aligned}$$

The random walk: mathematical analysis

Statistical independence

$$\Rightarrow \langle v_x(t - m\Delta t)v_x(t - n\Delta t) \rangle = 0, m \neq n$$

And for $m = n$

$$\langle v_x^2 \rangle = \frac{1}{5} \left(\frac{\Delta x^2}{\Delta t^2} \right) + \frac{3}{5} \times 0 + \frac{1}{5} \left(\frac{\Delta x^2}{\Delta t^2} \right) = \frac{2\Delta x^2}{5\Delta t^2}$$

The random walk: mathematical analysis

The sum can be separated in two terms

$$d_x^2 = \Delta t^2 \left(\sum_{\substack{m, n = 1 \\ m \neq n}}^N \langle v_x(t - m\Delta t) v_x(t - n\Delta t) \rangle + \sum_{m=1}^N \langle v_x(t - m\Delta t)^2 \rangle \right)$$
$$= \sum_{m=1}^N \frac{2\Delta x^2}{5} = N \frac{2\Delta x^2}{5} = t \frac{2\Delta x^2}{5\Delta t}$$

The random walk: mathematical analysis

Doing similarly for d_y^2 on gets

$$d = \sqrt{d_x^2 + d_y^2} = \sqrt{t \frac{4\Delta x^2}{5\Delta t}}$$

Finally by defining the diffusion coefficient $D \equiv \frac{\Delta x^2}{5\Delta t}$

$$d = \sqrt{4Dt}$$

Equivalent to

$$\partial_t \rho(x, y, t) = D(\partial_x^2 + \partial_y^2) \rho(x, y, t)$$

The random walk: 2D diffusion

Parameters of the simulation

$$t = 100, \quad \Delta x = \Delta t = 1, \quad (x_0, y_0) = (0, 0).$$

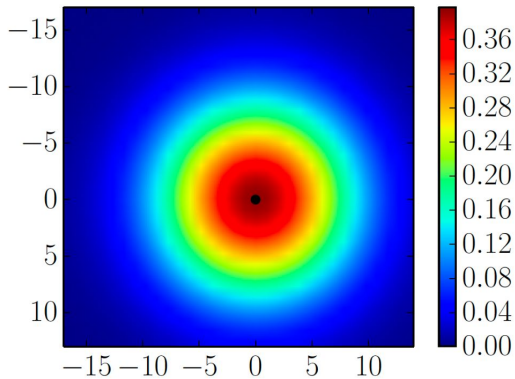
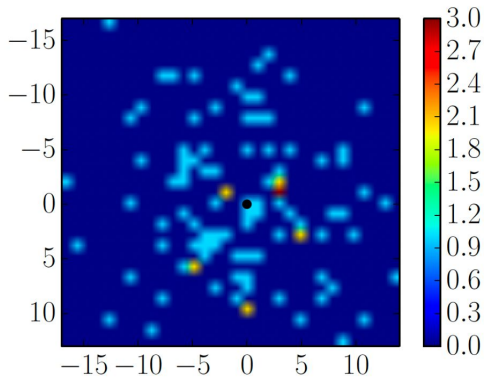
Transport parameters

$$D = \frac{1}{5}, \quad d = \sqrt{80} \cong 8.9443$$

Let us measure the average position and the average distance at time t for a different number of walkers.

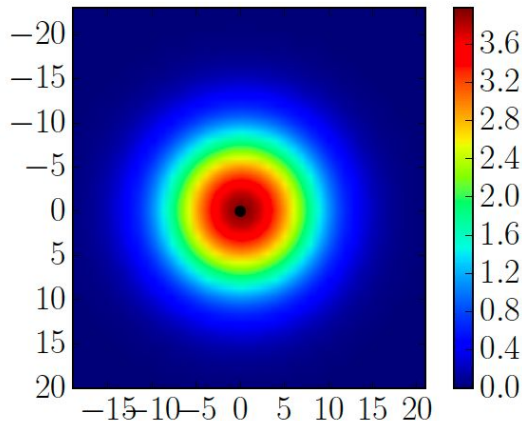
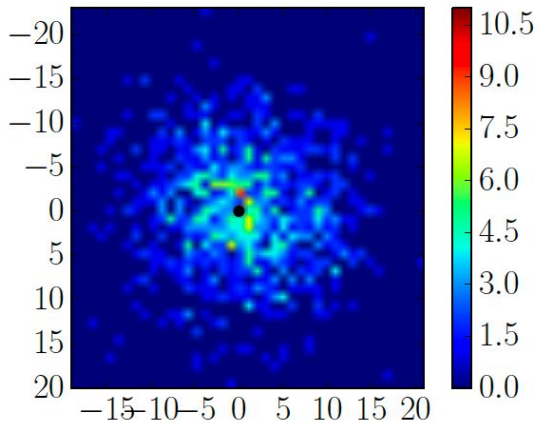
The random walk: $N = 100$

$$\langle (x, y) \rangle = (-0.63, -0.01), \quad d = 9.262$$



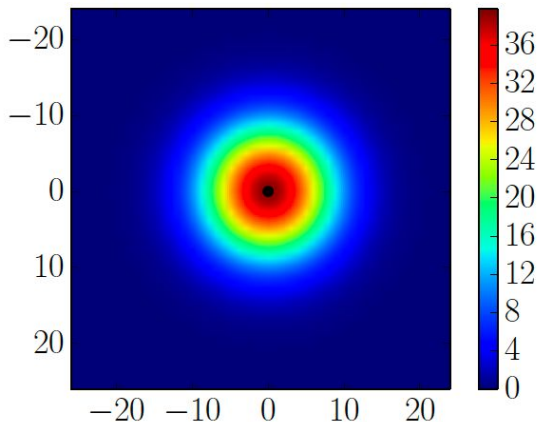
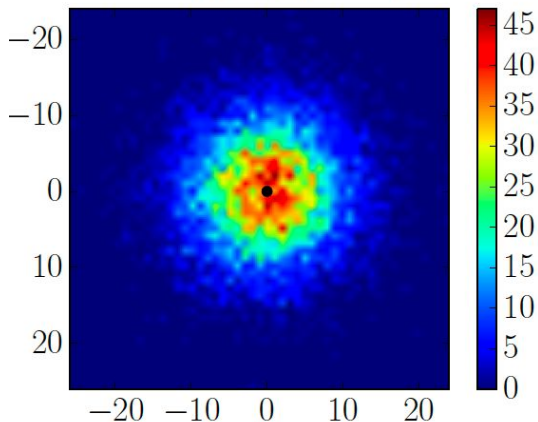
The random walk: $N = 1000$

$$\langle (x, y) \rangle = (0.382, -0.094), \quad d = 9.037$$



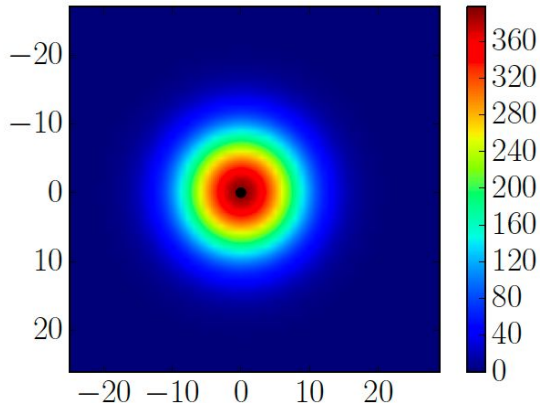
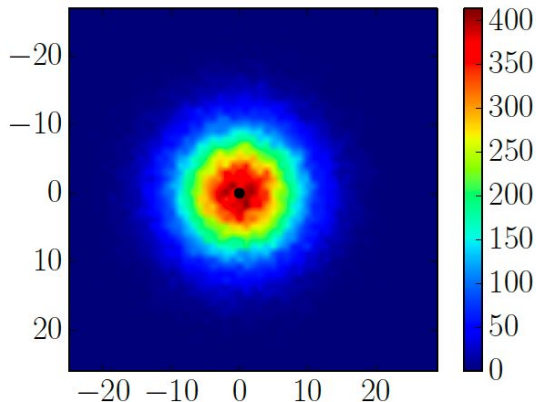
The random walk: $N = 10000$

$$\langle (x, y) \rangle = (0.0034, 0.0335), \quad d = 8.972$$



The random walk: $N = 100000$

$$\langle (x, y) \rangle = (-0.00214, -0.0013), \quad d = 8.938$$



3. Growth of a population

Growth of a population

- Let us note the number of individuals of a population $P(t)$, and $P(0) = P_0$.
- Only two processes are affecting the number of individuals: the number of births $B(t)$ and the number of deaths $D(t)$.
- The variation of $P(t)$ can be expressed as

$$\dot{P}(t) = B(t) - D(t).$$

- A possible model for births and deaths is given by

$$B(t) = r_b P(t), \quad D(t) = r_d P(t).$$

Growth of a population

Total variation of the population

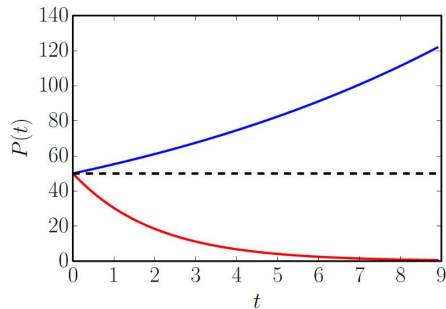
$$\dot{P}(t) = rP(t), \quad r = r_b - r_d.$$

The solution of this equation

$$P(t) = P_0 e^{rt}.$$

Growth of a population

- $r > 0$ exponential growth.
- $r < 0$ exponential decay.
- $r = 0$ constant.



Growth of a population

- Of course an infinite growth of population is not realistic.
- One can add a target “capacity” C to the population by $r \rightarrow 0$ as $P \rightarrow C$

$$r = v \left(1 - \frac{P}{C} \right).$$

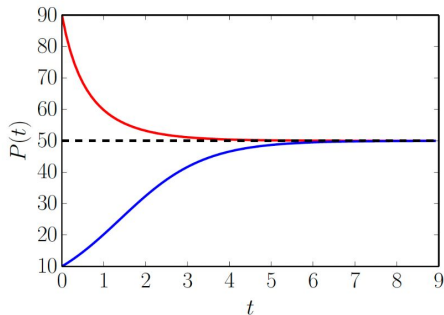
- The model becomes

$$\dot{P} = v \left(1 - \frac{P}{C} \right) P$$

Growth of a population

The solution is a logistic function

$$P(t) = \frac{C}{1 + \frac{C-P_0}{P_0}e^{-vt}}$$



$C = 50$, $v = 1$, $P_0 = 10$ (blue), 90 (red).

Growth of a population

- $P_0 < C$ growth of population and $P \rightarrow C$.
- $P_0 > C$ decay of population and $P \rightarrow C$.
- Except if $P_0 = 0$ then $P(t) = 0$.

This system has two fixed points: C (attractive) and 0 (repulsive).

4. Balance equations I

Balance equations

As shown with the growth of a population example the basic idea is to write a **balance equation**

$$\underbrace{\text{variation of a quantity}}_{\text{ex: } \dot{P}} = \underbrace{\text{creation rate}}_{\text{ex: } B(t)} - \underbrace{\text{destruction rate}}_{\text{ex: } D(t)}.$$

Or to reuse the generic formalism of continuous dynamical systems

$$\dot{s} = f(s) = \text{creation rate} - \text{destruction rate}.$$

Balance equations

For discrete dynamical systems this equation becomes

$$s(t + \Delta t) - s(t) = \Delta t (\text{creation rate} - \text{destruction rate}) .$$

Both formulations are equivalent in the limit $\Delta t \rightarrow 0$ since

$$\lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t} = \dot{s}.$$

Discrete balance equations can be directly implemented on a computer.

Lotka–Volterra model (“Prey-Predator”)

Let a be the antelope population and c the cheetah population.

$$\frac{da}{dt} = \underbrace{k_a a(t)}_{(1)} - \underbrace{k_{c,a} c(t) a(t)}_{(2)},$$

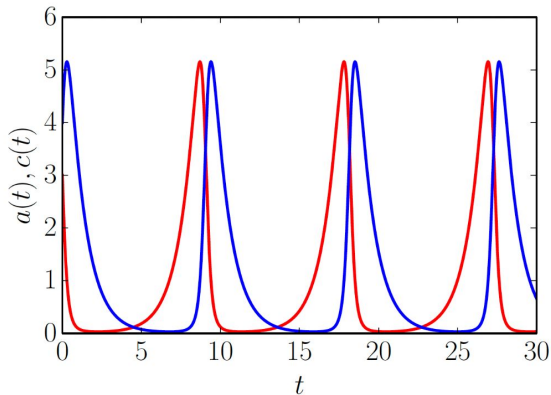
$$\frac{dc}{dt} = -\underbrace{k_c c(t)}_{(3)} + \underbrace{k_{a,c} a(t) c(t)}_{(4)}$$

- The antelopes reproduce exponentially fast (infinite amount of food).
- The antelopes are being eaten by the cheetah proportionally to their chance to meet.
- The cheetahs die exponentially fast of starvation.
- The cheetahs are reproducing proportionally to their chance to eat antelopes.

Lotka–Volterra model (“Prey-Predator”)

Time evolution

Periodicity



Lotka–Volterra model (“Prey-Predator”)

Parametric plot

Initial condition
sensitivity

